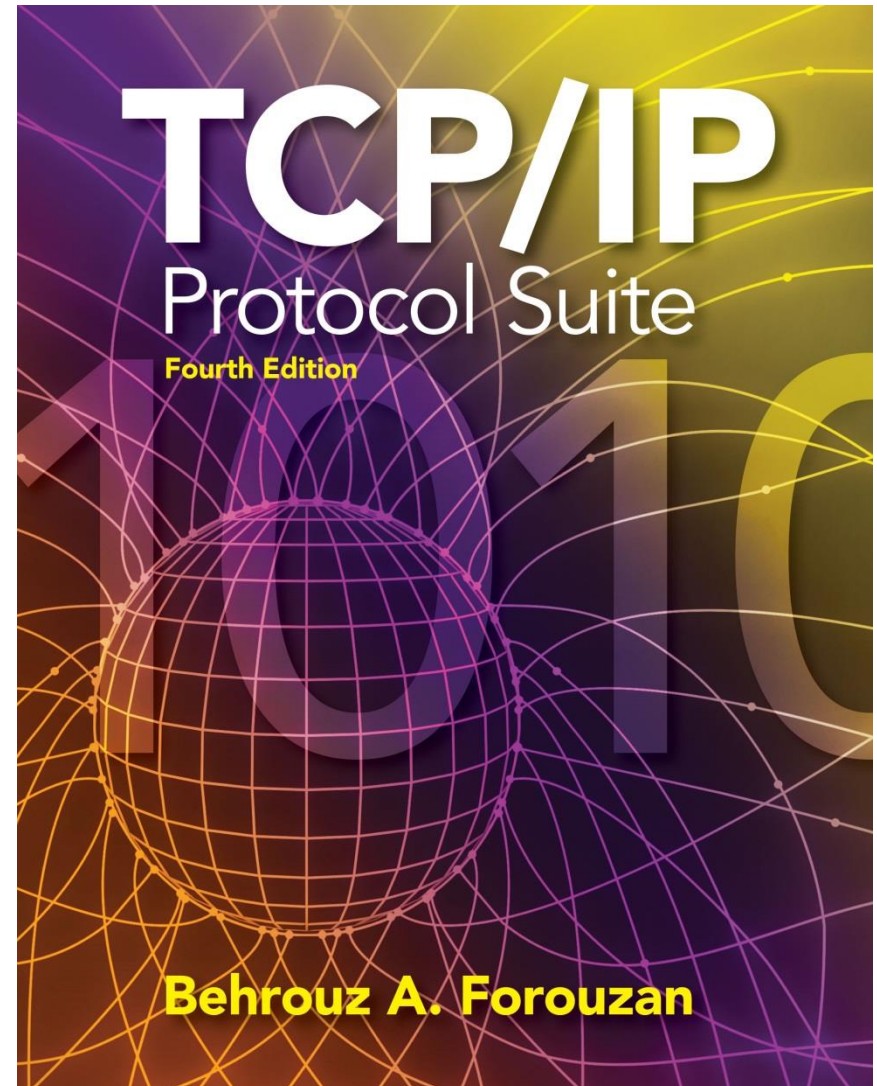


Chapter 6

Delivery and Forwarding of IP Packets



OBJECTIVES:

- ❑ To discuss the delivery of packets in the network layer and distinguish between direct and indirect delivery.
- ❑ To discuss the forwarding of packets in the network layer and distinguish between destination-address-based forwarding and label-based forwarding.
- ❑ To discuss different forwarding techniques, including next-hop, network-specific, host-specific, and default.
- ❑ To discuss the contents of routing tables in classful and classless addressing and some algorithms used to search the tables.

Chapter Outline

6.1 Delivery

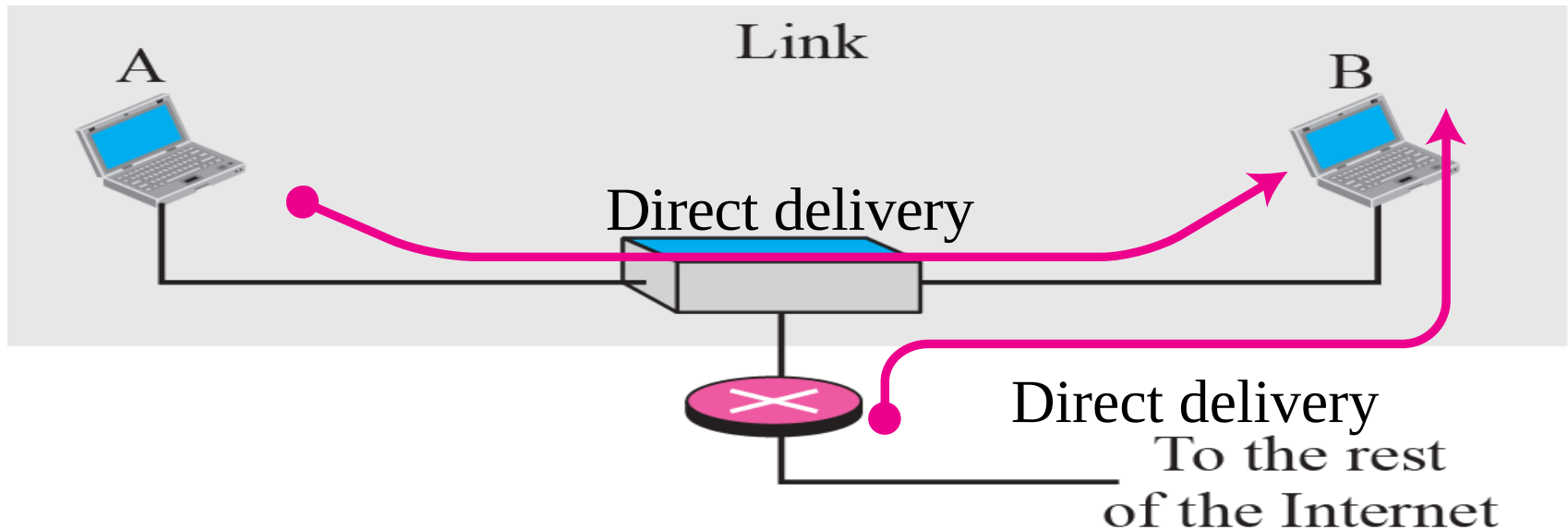
6.2 Forwarding

6-1 DELIVERY

The network layer supervises the handling of the packets by the underlying physical networks. We define this handling as the delivery of a packet. The delivery of a packet to its final destination is accomplished using two different methods of delivery:

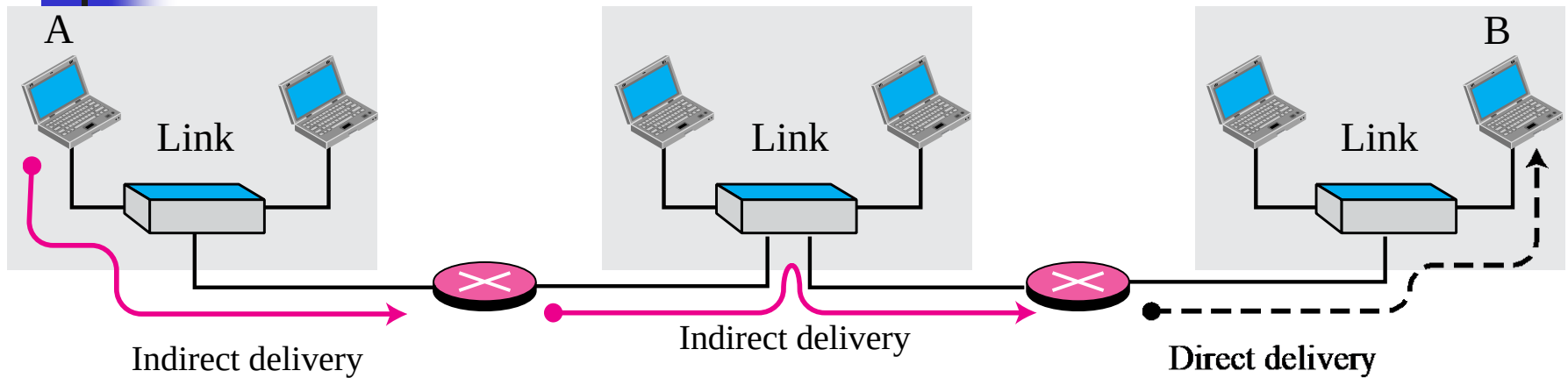
Direct and Indirect.

Figure 6.1 *Direct delivery*



In **direct Delivery**, the final destination of packet is a host connected to the same physical network as the deliverer.

Figure 6.2 *Indirect delivery*



In an **Indirect delivery**, the packet goes from router to router until it reaches the one connected to the same physical network as its final destination.

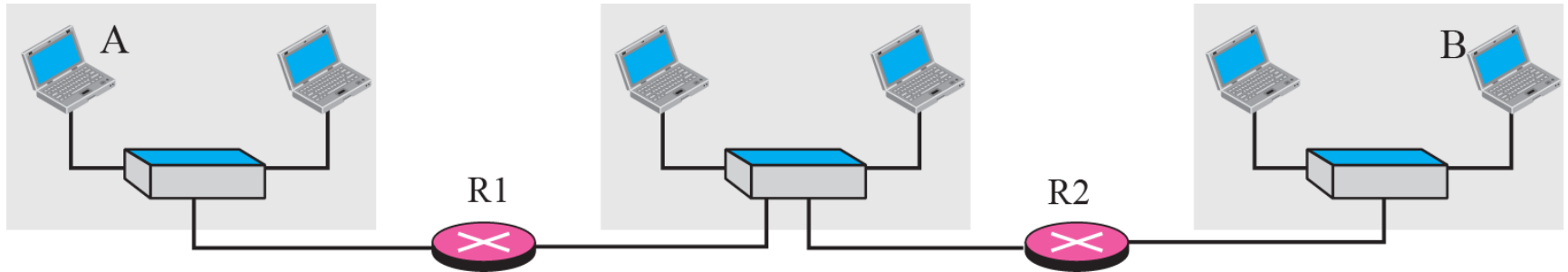
6-2 FORWARDING

Forwarding means to place the packet in its route to its destination. Since the Internet today is made of a combination of links (networks), forwarding means to deliver the packet to the next hop (which can be the final destination or the intermediate connecting device). Although the IP protocol was originally designed as a connectionless protocol, today the tendency is to use IP as a connection-oriented protocol.

Topics Discussed in the Section

- ✓ **Forwarding Based on Destination Address**
 - **Next Hop method**
 - **Network Specific method**
 - **Host-Specific Method**
 - **Default method**
- ✓ **Forwarding Based on Label**

Figure 6.3 *Next-hop method*



A		R1		R2	
Destination	Route	Destination	Route	Destination	Route
Host B	R1, R2, Host B	Host B	R2, Host B	Host B	Host B

a. Routing tables based on route

A		R1		R2	
Destination	Next Hop	Destination	Next Hop	Destination	Next Hop
Host B	R1	Host B	R2	Host B	---

b. Routing tables based on next hop

Figure 6.4 *Network-specific method*

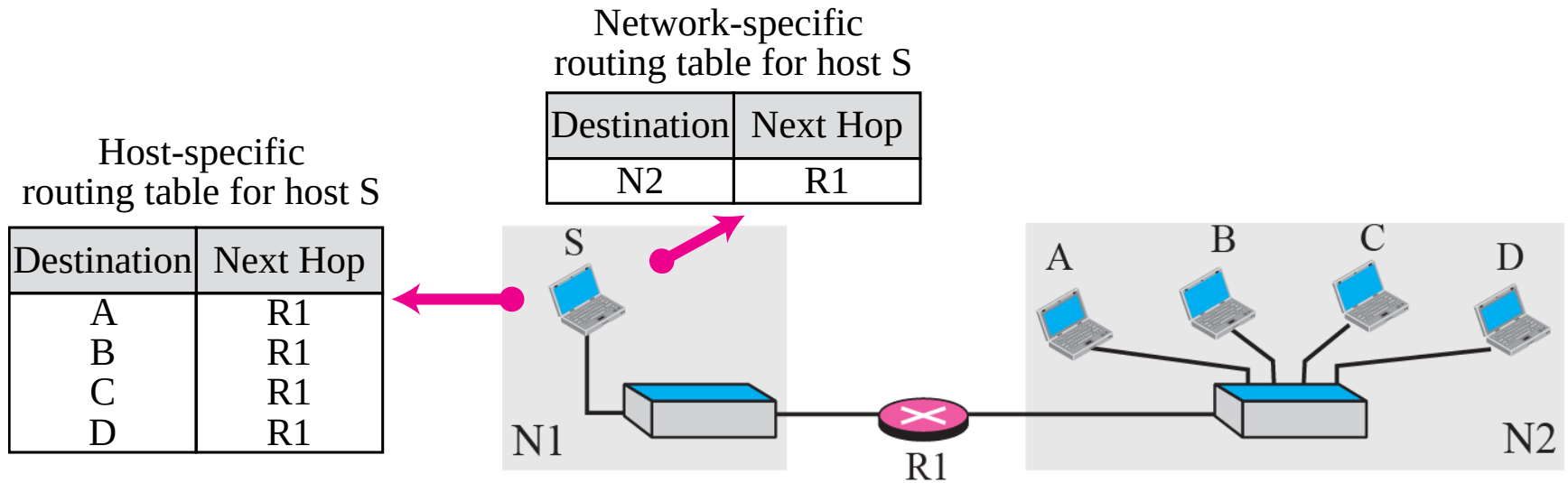


Figure 6.5 *Host-specific routing*

Routing table for host A

Destination	Next Hop
Host B	R3
N2	R1
N3	R3
.....	

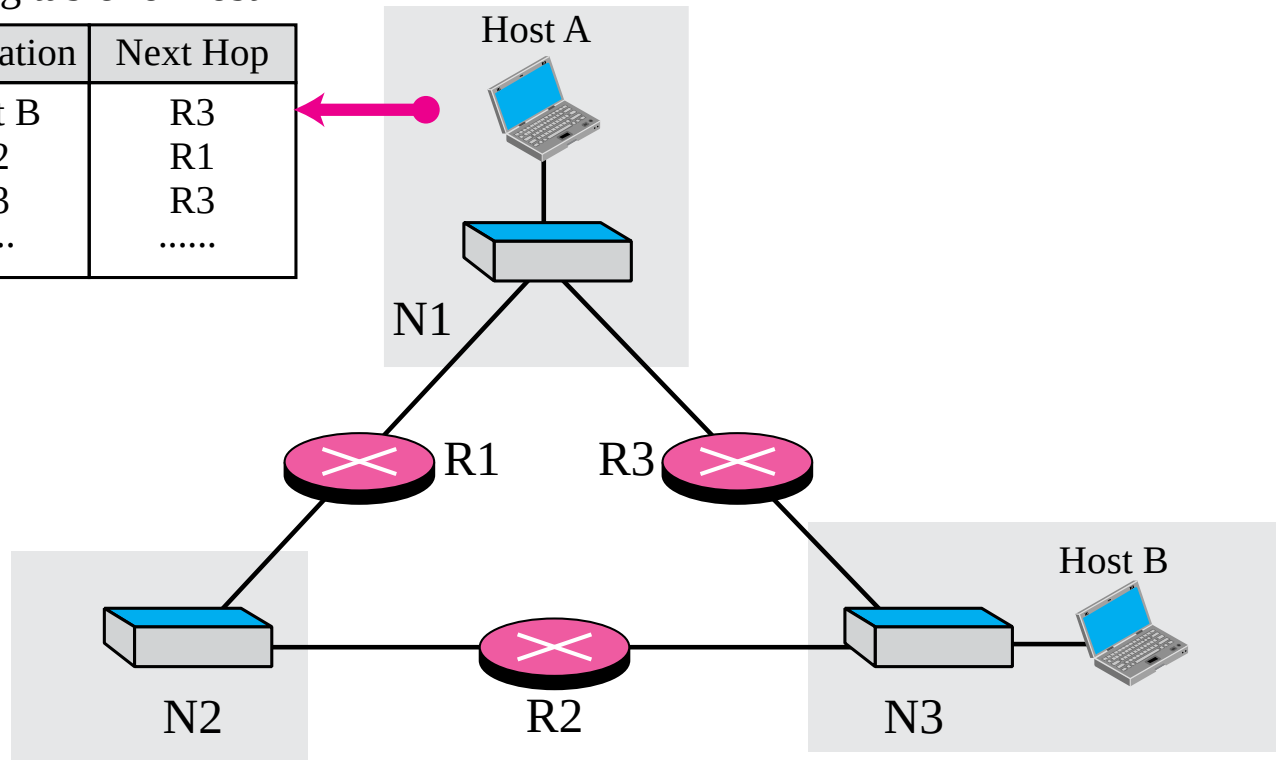


Figure 6.6 *Default routing*

Routing table for host A

Destination	Next Hop
N2	R1
.....	
Default	R2

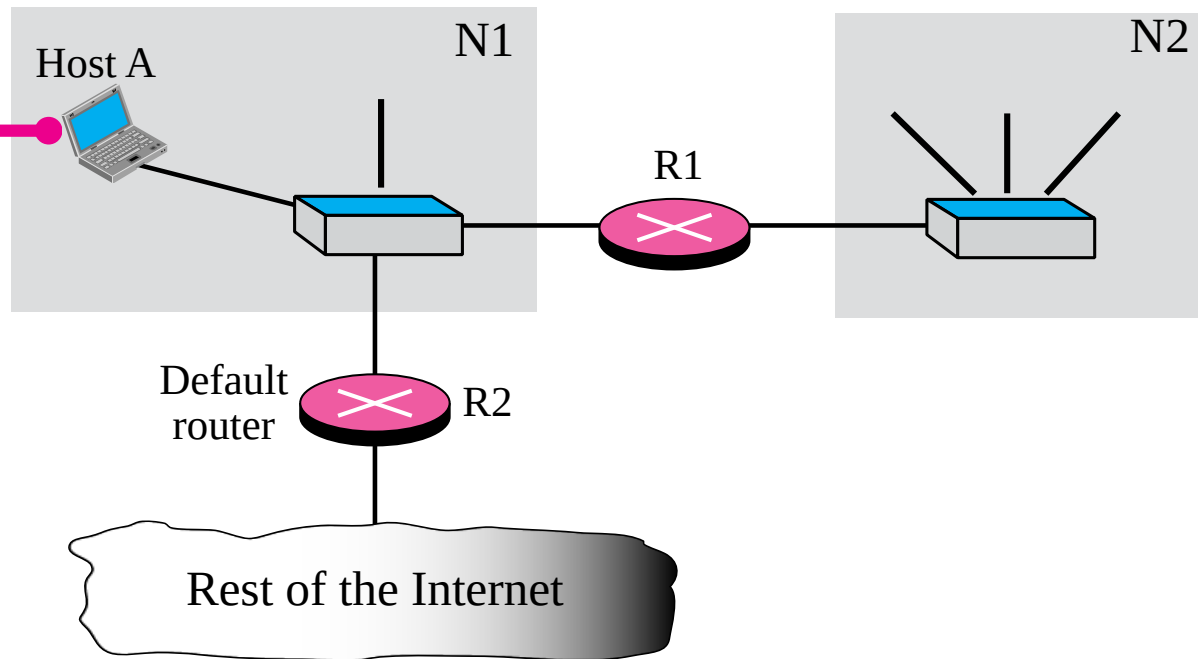
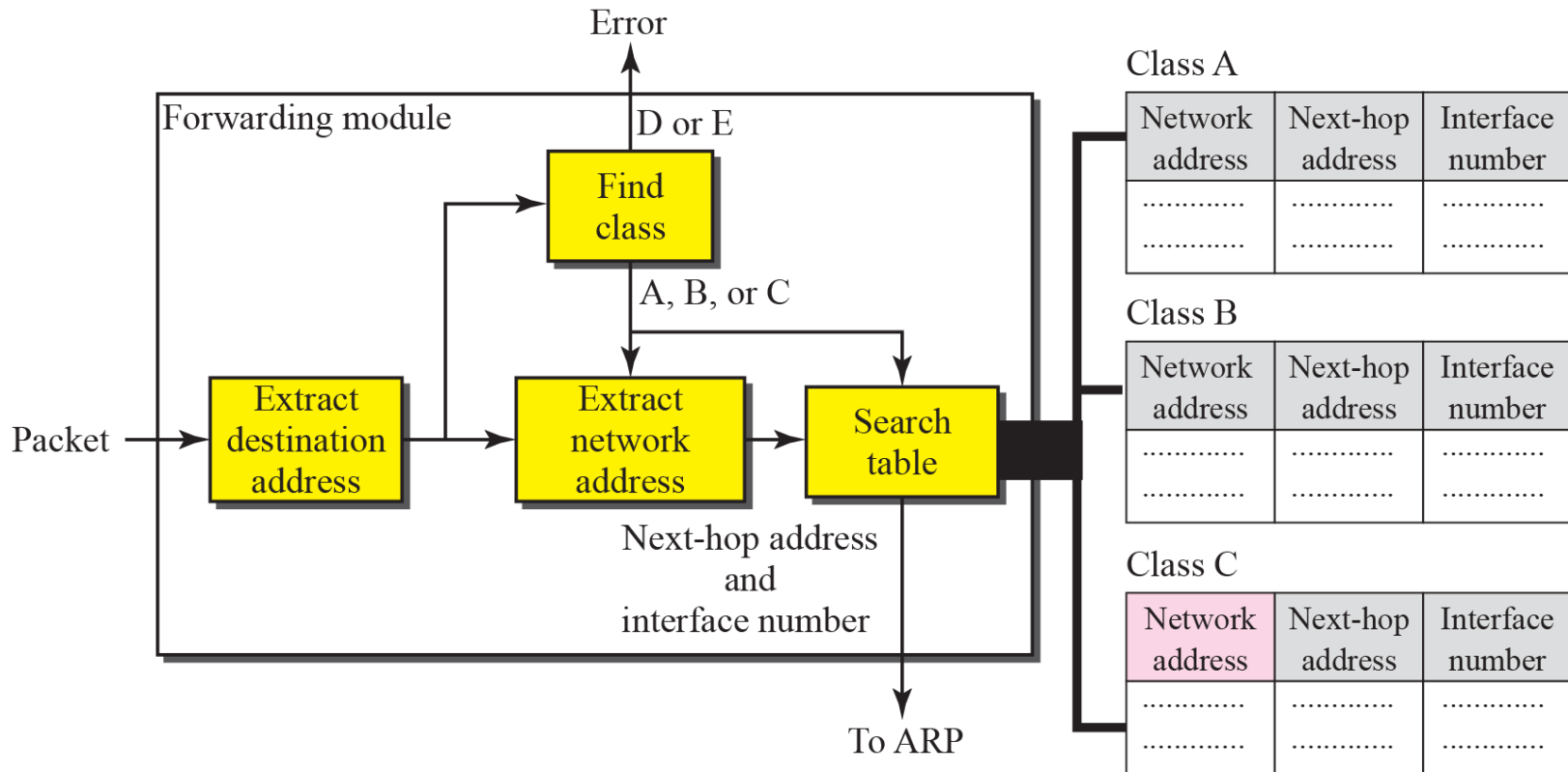


Figure 6.7 *Simplified forwarding module in classful address without subnetting*



Example 6.1

Figure 6.8 shows an imaginary part of the Internet. Show the routing tables for router R1.

Solution

Figure 6.9 shows the three tables used by router R1. Note that some entries in the next-hop address column are empty because in these cases, the destination is in the same network to which the router is connected (direct delivery). In these cases, the next-hop address used by ARP is simply the destination address of the packet as we will see in Chapter 8.

Figure 6.8 Configuration for routing, Example 6.1

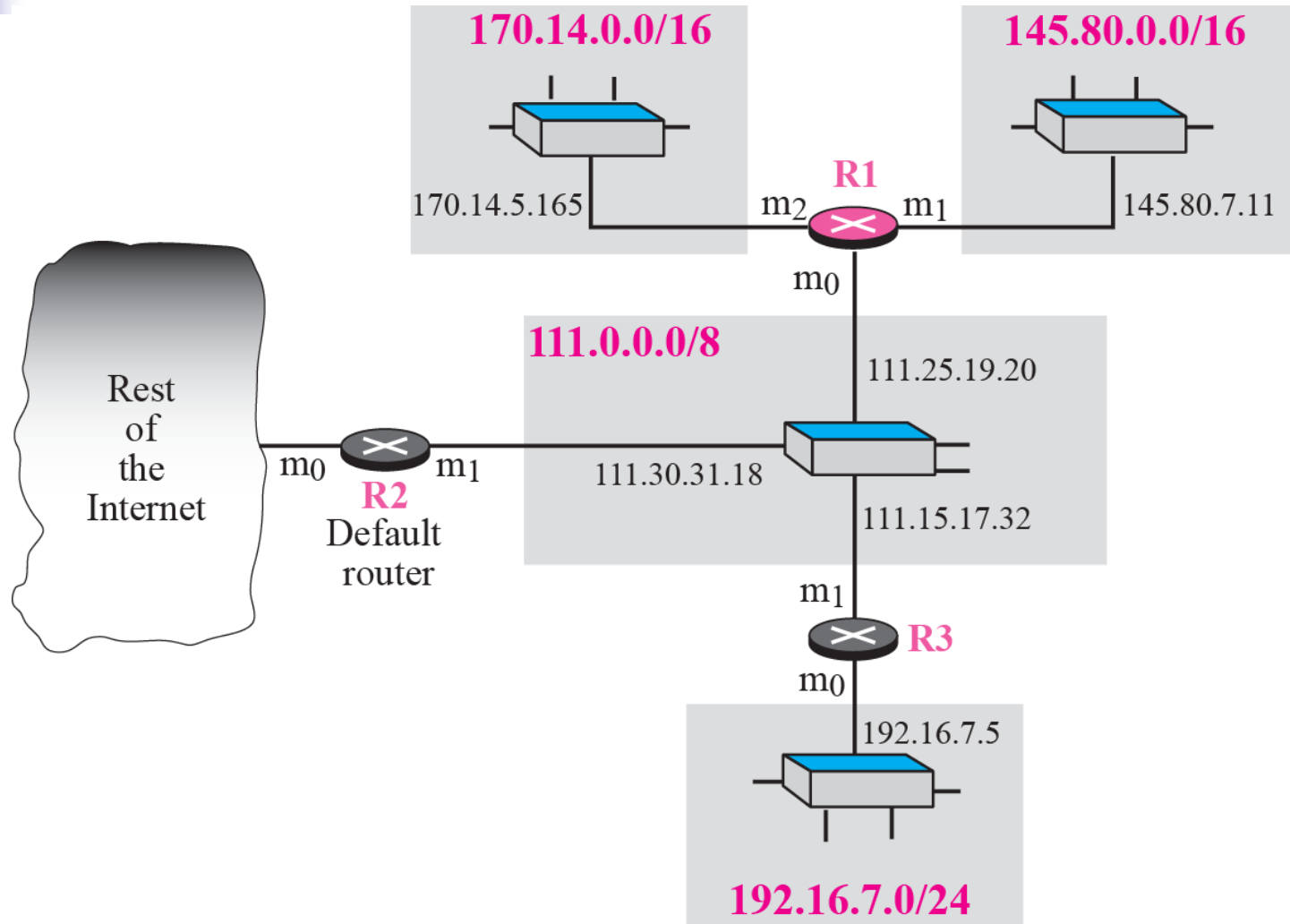




Figure 6.9 *Tables for Example 6.1*

Class A

Network address	Next-hop address	Interface
111.0.0.0	-----	m0

Class C

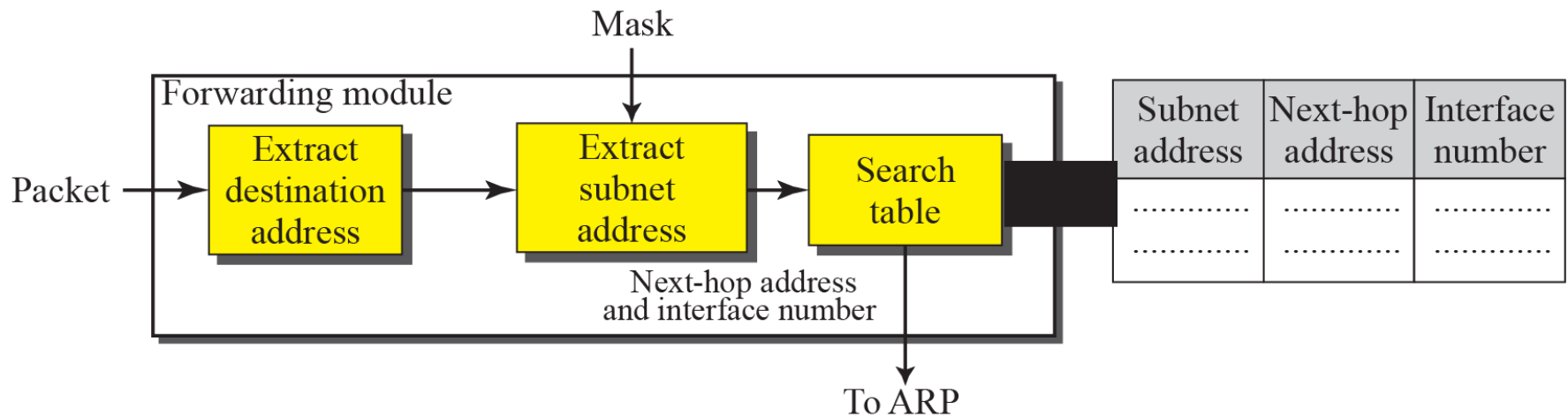
Network address	Next-hop address	Interface
192.16.7.0	111.15.17.32	m0

Class B

Network address	Next-hop address	Interface
145.80.0.0	-----	m1
170.14.0.0	-----	m2

Default: 111.30.31.18, m0

Figure 6.10 *Simplified forwarding module in classful address with subnetting*

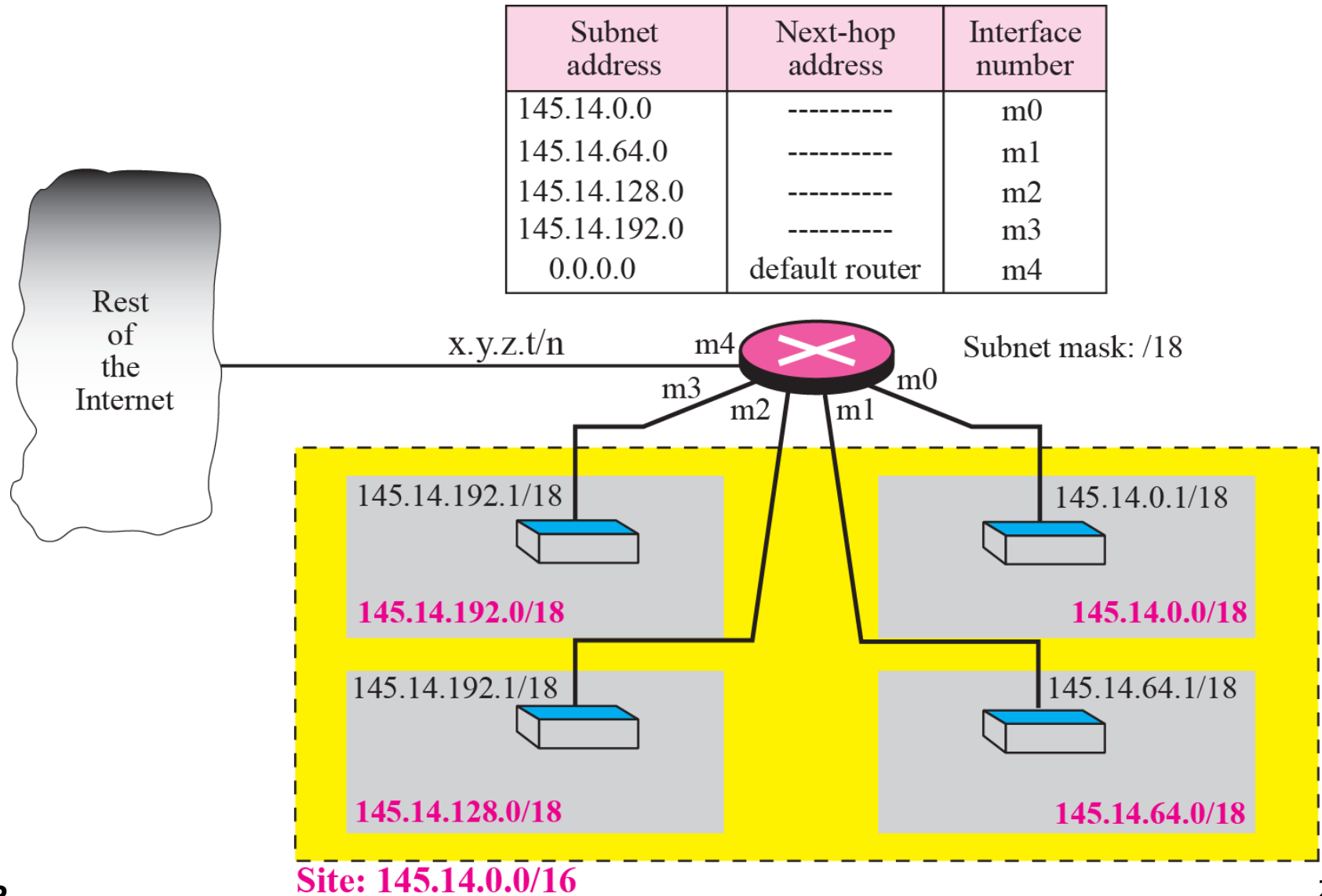


Example 6.4

Figure 6.11 shows a router connected to four subnets.

Note several points. First, the site address is 145.14.0.0/16 (a class B address). Every packet with destination address in the range 145.14.0.0 to 145.14.255.255 is delivered to the interface m4 and distributed to the final destination subnet by the router. Second, we have used the address x.y.z.t/n for the interface m4 because we do not know to which network this router is connected. Third, the table has a default entry for packets that are to be sent out of the site. The router is configured to apply the subnet mask /18 to any destination address.

Figure 6.11 Configuration for Example 6.4



Example 6.5

The router in Figure 6.11 receives a packet with destination address 145.14.32.78. Show how the packet is forwarded.

Solution

The mask is /18. After applying the mask, the subnet address is 145.14.0.0. The packet is delivered to ARP (see Chapter 8) with the next-hop address 145.14.32.78 and the outgoing interface m0.

Example 6.6

A host in network 145.14.0.0 in Figure 6.11 has a packet to send to the host with address 7.22.67.91. Show how the packet is routed.

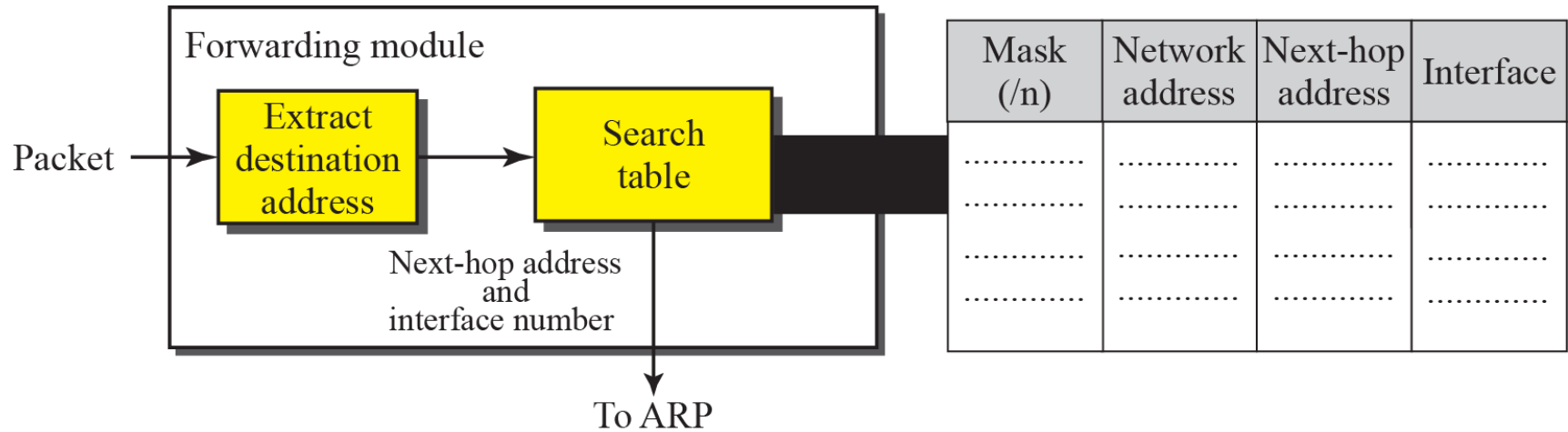
Solution

The router receives the packet and applies the mask (/18). The network address is 7.22.64.0. The table is searched and the address is not found. The router uses the address of the default router (not shown in figure) and sends the packet to that router.

Note

In classful addressing we can have a routing table with three columns; in classless addressing, we need at least four columns.

Figure 6.12 *Simplified forwarding module in classless address*



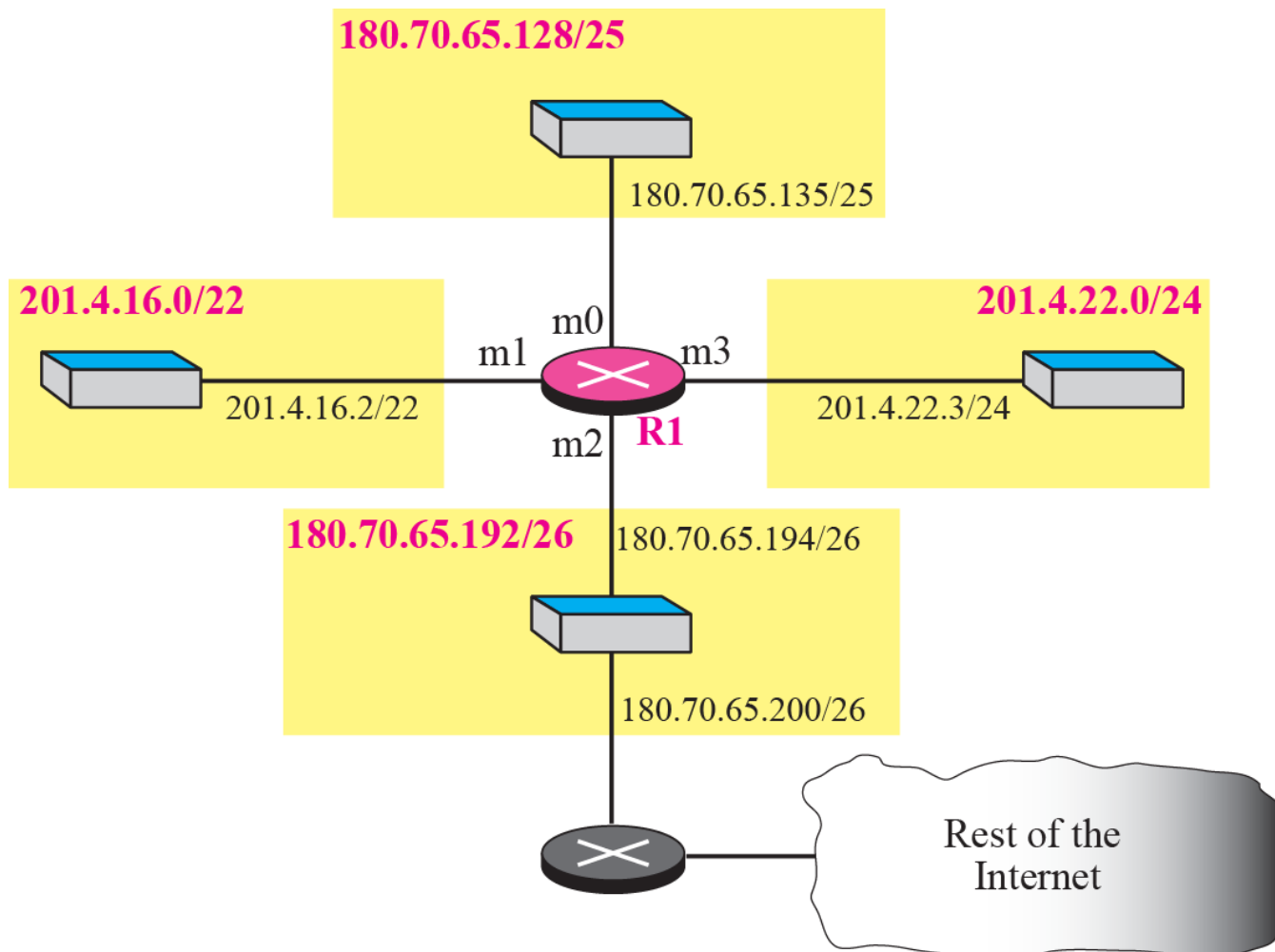
Example 6.7

Make a routing table for router R1 using the configuration in Figure 6.13.

Solution

Table 6.1 shows the corresponding table

Figure 6.13 *Configuration for Example 6.7*



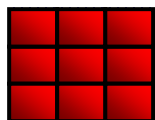


Table 6.1 *Routing table for router R1 in Figure 6.13*

<i>Mask</i>	<i>Network Address</i>	<i>Next Hop</i>	<i>Interface</i>
/26	180.70.65.192	-	m2
/25	180.70.65.128	-	m0
/24	201.4.22.0	-	m3
/22	201.4.16.0		m1
Default	Default	180.70.65.200	m2

Example 6.8

Show the forwarding process if a packet arrives at R1 in Figure 6.13 with the destination address 180.70.65.140.

Solution

The router performs the following steps:

1. The first mask (/26) is applied to the destination address. The result is 180.70.65.128, which does not match the corresponding network address.
2. The second mask (/25) is applied to the destination address. The result is 180.70.65.128, which matches the corresponding network address. The next-hop address (the destination address of the packet in this case) and the interface number m0 are passed to ARP for further processing.

Example 6.9

Show the forwarding process if a packet arrives at R1 in Figure 6.13 with the destination address 201.4.22.35.

Solution

The router performs the following steps:

- 1.** The first mask (/26) is applied to the destination address. The result is 201.4.22.0, which does not match the corresponding network address (row 1).
- 2.** The second mask (/25) is applied to the destination address. The result is 201.4.22.0, which does not match the corresponding network address (row 2).
- 3.** The third mask (/24) is applied to the destination address. The result is 201.4.22.0, which matches the corresponding network address.

Example 6.10

Show the forwarding process if a packet arrives at R1 in Figure 6.13 with the destination address 18.24.32.78.

Solution

This time all masks are applied to the destination address, but no matching network address is found. When it reaches the end of the table, the module gives the next-hop address 180.70.65.200 and interface number m2 to ARP (see Chapter 8). This is probably an outgoing package that needs to be sent, via the default router, to someplace else in the Internet.

Example 6.11

Now let us give a different type of example. Can we find the configuration of a router if we know only its routing table? The routing table for router R1 is given in Table 6.2. Can we draw its topology?

Solution

We know some facts but we don't have all for a definite topology. We know that router R1 has three interfaces: m0, m1, and m2. We know that there are three networks directly connected to router R1. We know that there are two networks indirectly connected to R1. There must be at least three other routers involved (see next-hop column). We do not know if network 140.6.12.64 is connected to router R3 directly or through a point-to-point network (WAN) and another router. Figure 6.14 shows our guessed topology.

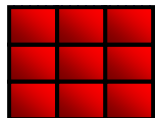


Table 6.2 *Routing table for Example 6.11*

<i>Mask</i>	<i>Network Address</i>	<i>Next-Hop Address</i>	<i>Interface Number</i>
/26	140.6.12.64	180.14.2.5	m2
/24	130.4.8.0	190.17.6.2	m1
/16	110.70.0.0	-----	m0
/16	180.14.0.0	-----	m2
/16	190.17.0.0	-----	m1
Default	Default	110.70.4.6	m0